

Effects of a handlebar on standing VR locomotion

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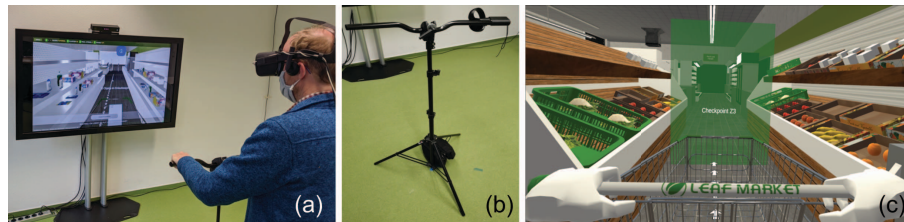


Figure 1: User performing walk in place locomotion (a) with a handlebar (b) in a VR supermarket environment (c).

ABSTRACT

To provide a natural locomotion for a virtual supermarket in a cognitive rehabilitation application we compare three walk in place (WIP) approaches (WIP; WIP + Bar for stability support; WIP + Bar for steering and stability support). We use tracked knee bending with and without a handlebar as a real correspondent to a virtual handlebar of a virtual shopping cart. For steering we use shoulder or handlebar rotation. We present an evaluation of the influence of the physical handlebar as a fixed reference and postural stability aid on cybersickness and user satisfaction. Our preliminary results indicate advantages for steering via a handlebar. Unexpectedly, a bar for postural stability seems to increase cybersickness but still improve user satisfaction compared to WIP only.

Keywords: VR locomotion, VR navigation, cybersickness, handlebar, postural stability, grounding

Index Terms: Human-centered computing → Virtual reality

1 INTRODUCTION

VR becomes popular in neurological and rehabilitation assessments and exercises for controlled simulation of complex environments that are difficult to setup physically in a laboratory. For such tasks, VR systems have to meet higher requirements than for an entertainment setup. The system needs to be suitable for sensitive/restricted users without limiting their VR interactions. To minimize the difference to the real-world interaction fidelity is desirable. So besides finding alternative metaphors for natural interaction, versatile research is done to enable natural interaction in VR [4, 5, 9]. For lack of better natural locomotion alternatives, teleportation is currently the most commonly used locomotion mechanism in VR. However, teleportation is not suitable for VR applications simulating reality, e.g., in search tasks during neurological tests.

The implementation of locomotion is still a major challenge in virtual reality applications as it causes or amplifies cybersickness. This is why measurement and reduction of cybersickness-

influencing factors are being studied intensively [7]. To overcome the decoupling of body movement and navigation with classical input devices e.g., mouse/keyboard, gamepad, touchscreen, and to be able to use the hands for interaction in VR instead of using them for locomotion, the focus is on more natural locomotion and navigation. Guy et al. [1] e.g., proposed “LazyNav”, a mid-air ground navigation control model making use of a single pair of tracked body elements to tailor the navigation. Although they did not find an ideal combination of body interaction for rotating and moving, rotation of the upper body part (bust, shoulder) turned out to be a good metaphor for rotation. For the walk interaction, stepping, bending knees and bending the bust all had their advantages and drawbacks. Punpongson [5] translated these findings to HDMs and showed that the results for locomotion in front of a large display cannot be directly transferred to HMDs.

Besides cybersickness there are more issues with WIP implementations in real-world usage scenarios. Moving feet, even if intended in one place, can cause a drift in position. Some authors reported that they bypassed this user problem by warnings from the experimenter, during the test trials [9]. Offering something to lean on (e.g., a chair) can help to maintain balance [4].

The specific locomotion mode is also influenced by the technology used for tracking the body movement. Body worn sensors (e.g., shoes with pressure sensor), external cameras or only the HMD included sensors can be used. Our technological developments focus on camera-based, contact-free spatial tracking of user movements since it offers flexibility and ease of use in any kind of movement tracking.

In accordance with the postural instability theory [6] and the need for natural VR interaction, we propose a Walk In Place (WIP) [1] approach using knee bending as sensory input for forward movement. In this study we evaluate WIP locomotion with handlebars to examine effects on postural stability, cybersickness and user satisfaction.

2 SYSTEM DESCRIPTION

Inspired by our research project VReha [8], we implemented a virtual supermarket as a testbed in Unity and displayed it on an Oculus Rift HMD. The scene complexity was adjusted to guarantee a 90Hz frame rate. The aisles in the virtual supermarket are approx. 1.5 m wide, resembling common real supermarkets in Germany.

To support postural stability, we mounted a handlebar on a tripod. This construction is simple, inexpensive, adaptable to various user dimensions and represents the real counterpart to the virtual handlebar of the shopping trolley. One Oculus Rift controller was mounted to the handlebar to determine its position

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and track its rotation using the VR system's built-in controller tracking (see Fig. 1). This allows us to map the virtual handlebar to the real handlebar for an accurate representation within the virtual environment, which leads to a fixed reference between the real and virtual world. Besides a shopping cart we can imagine various other metaphors for the use of a handlebar, such as a bicycle, e-shopper, Segway, scooter, rollator or a pushchair.

The detection of shoulder rotation and knee bending was implemented based on the skeleton detection from the Microsoft Kinect v2 camera. A shoulder rotation of more than 10° was translated into direction changes. In WIP, knee bending detection tolerates active walking and setting one foot in front of the other as well. Tracking is done at 30Hz. Due to filtering the latency is slightly bigger.

3 METHOD

We compare the following locomotion techniques (LT):

- *1_OnlyWIP*: WIP + shoulder rotation
- *2_BarWIP*: WIP + shoulder rotation + a bar as physical stability support and real world anchor, without steering functionality
- *3_HandleBarWIP*: WIP + a handlebar as physical stability support, with VR reference and steering functionality

The task was to locomote in the supermarket following arrows displayed on the ground. The whole path was divided into 30 checkpoints. Travel-time between checkpoints was measured. Travel speed was the same for all LTs. To familiarize with the navigation and rotation techniques, all users had to walk one round path in the supermarket prior to the actual experiment. Each trial took in total approximately 300 seconds per LT.

Before and after each trial, test subjects answered the Simulator Sickness Questionnaire (SSQ) [2]. During the trial, at every fifth checkpoint (FMS-checkpoint), they reported their cybersickness using the Fast Motion Sickness Scale (FMS), a verbal rating scale ranging from zero (no sickness at all) to 20 (frank sickness) [3]. After each trial users had to answer questions (7-point Likert scale) concerning their ability to move forward and change direction, their perceived stability, their reference to the real world and their overall satisfaction (7-point Kunin scale) for the tested motion control.

Each test subject performed all trial conditions in a randomized order. Participants with sickness symptoms paused between each test session until cybersickness recovery.

Due to the COVID-19 pandemic only six internal users could be evaluated. All participants were instructed prior to each trial but had no supervision during the experiment to avoid contact.

4 RESULTS

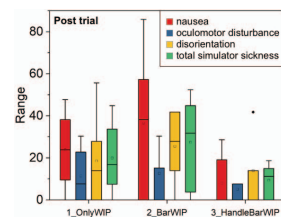


Figure 2: SSQ after the trial

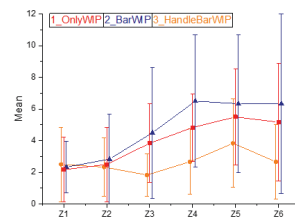


Figure 3: Mean FMS values at FMS-checkpoints per condition

In condition *1_OnlyWIP* two drop outs happened because participants drifted out of the tracking area of the camera.

It turned out, that *3_HandleBarWIP* performs best in terms of cybersickness (see Fig. 2 and Fig. 3).

No differences could be found concerning the travel-time between checkpoints. Overall satisfaction with motion control (7-point Kunin scale) was judged best for *3_HandleBarWIP* ($M = 5.7$), followed by *2_BarWIP* ($M = 4.7$) and *1_OnlyWIP* ($M = 3.7$).

Both conditions with a bar got better ratings for postural stability than the WIP only condition (mean values: 6.5 and 5.8 versus 4.0). Even though forward locomotion was implemented in the same way for all conditions via WIP, differences were found in the assessment of the ability to control acceleration and deceleration (mean values: *3_HandleBarWIP*: 5.7, *2_BarWIP*: 5.8, *1_OnlyWIP*: 3.8). The fewest problems with left-right rotation were reported for *3_HandleBarWIP* ($M = 1.3$), followed by *2_BarWIP* ($M = 3.3$), and *1_OnlyWIP* ($M = 4.1$). No differences were found according their assessment of having a reference to the real world.

5 DISCUSSION AND CONCLUSION

It is not surprising that the support bar we provided contributes to a safer hold, more stability in VR and thus higher user satisfaction. We expected cybersickness reduction with this postural stability support, which turned out to be true for condition *3_HandleBarWIP*. Unexpectedly *2_BarWIP* caused the highest scores for cybersickness, even more than *1_OnlyWIP*. A possible explanation for the increased cybersickness in this condition is a sensory conflict between the visual and haptic experience, as the rotation of the real bar has no influence on the VR visualization. Surprisingly, the increased cybersickness did not affect the overall user satisfaction with this condition. We could identify a clear ranking, from the most satisfactory *3_HandleBarWIP* to *2_BarWIP* and the worst rated *1_OnlyWIP* locomotion technique.

The handlebar proved to be beneficial for rotation control in VR and lower cybersickness scores, but without advantages in efficiency as the test subjects were not faster.

Due to the pandemic, we could only test limited subjects and will extend our study to include a larger sample size to confirm our preliminary results. We also plan to examine the sensory conflict theory for LT *2_BarWIP* and an implementation of adjustable travel speed and longer locomotion.

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